

ReconWraith

Automated Reconnaissance Tool for Linux

Project Report

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Classification: Internal / Authorized Security Testing Use Only

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1. Project Overview

ReconWraith is a lightweight, single-file, automated reconnaissance tool built for Linux environments. It was designed to consolidate the repetitive early-stage steps of an authorized penetration test or bug bounty engagement — WHOIS lookup, DNS enumeration, subdomain discovery, HTTP header/technology fingerprinting, and port scanning — into a single command that produces a clean, shareable report.

The tool is written in pure Python 3 with no mandatory third-party dependencies. Where system security tools such as `nmap`, `whois`, and `dig` are available, ReconWraith automatically uses them for richer results; where they are not installed, it falls back to built-in equivalents (a threaded socket-based port scanner, Python's socket resolver for DNS, etc.) so the tool still functions on a minimal system.

1.1 Goals

- Automate the repetitive first steps of authorized recon so analysts spend less time on manual lookups.
- Work out-of-the-box on any Linux box with Python 3, with graceful fallbacks when optional tools are missing.
- Produce a clean, portable report (.txt and .json) suitable for attaching to an assessment deliverable.
- Enforce a clear authorization/legal-use checkpoint before every scan.

2. Legal & Ethical Use Statement

ReconWraith is intended strictly for use against systems the operator owns or is explicitly authorized in writing to test (e.g., a signed penetration testing agreement, an in-scope bug bounty program, or a personal lab). Unauthorized scanning of third-party systems may violate computer misuse laws in many jurisdictions. To reinforce this, the tool displays a mandatory authorization prompt before every scan unless explicitly bypassed with the `-y` flag, which is intended only for pre-approved, automated/CI use.

3. Tool Architecture

ReconWraith is organized into independent modules, each responsible for one recon function. A central driver resolves the target, then calls each enabled module in sequence, aggregating results into a single dictionary that is serialized to the final report.

3.1 Module Summary

Module	Description
WHOIS	Runs system <code>whois</code> binary if available; reports registrar, dates, name servers.
DNS Enumeration	Uses <code>dig</code> for A/AAAA/MX/NS/TXT/SOA/CNAME records, or falls back to a basic A-record lookup.
HTTP Header Grab	Connects to ports 80/443 and records status code and response headers.
Tech Fingerprint	Parses <code>Server</code> / <code>X-Powered-By</code> headers captured by the HTTP module.
Port Scan	Uses <code>nmap -sV -Pn</code> if installed; otherwise a threaded Python socket scanner across common ports.
Subdomain Enum	Queries crt.sh certificate transparency logs plus a built-in/custom wordlist brute force, resolving each candidate.
Robots/Sitemap (--full)	Fetches robots.txt and sitemap.xml over HTTP/HTTPS if reachable.

3.2 Data Flow

target input → DNS resolution → parallel module execution → results dictionary → report writer (txt / json) → output file(s).

4. Installation

ReconWraith requires only Python 3.8+. Optional system tools (nmap, whois, dnsutils) enhance results but are not required:

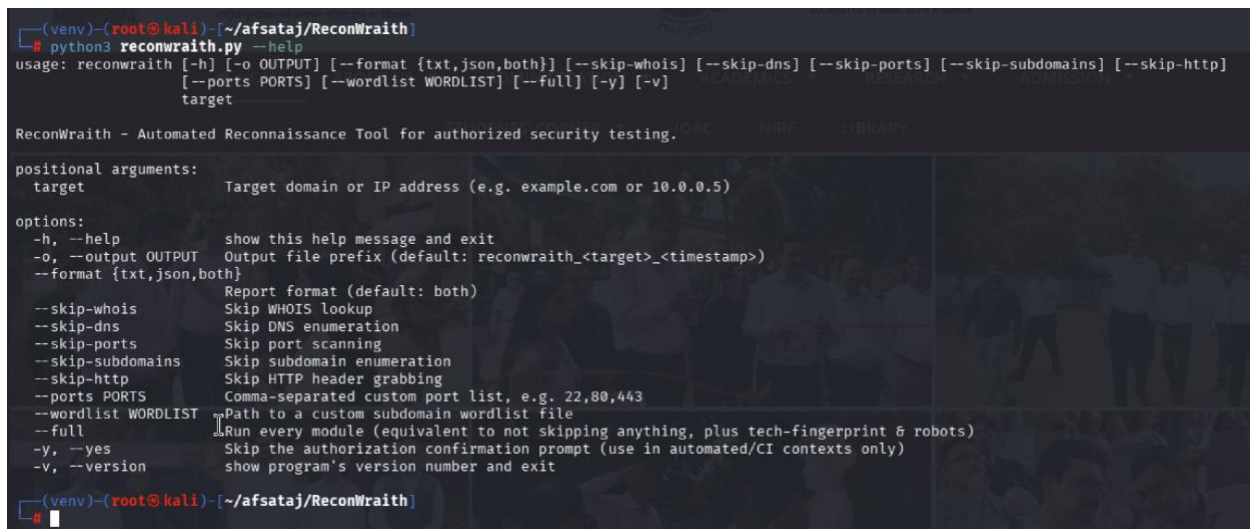
- Copy reconwraith.py to a working directory, e.g. ~/tools/reconwraith/
- Run: `chmod +x reconwraith.py`
- Optional: `sudo cp reconwraith.py /usr/local/bin/reconwraith` to add it to PATH
- Optional: `sudo apt install -y nmap whois dnsutils` for full-featured scans

5. Demonstration & Screenshots

This section documents the expected screenshots for this report. Each placeholder below indicates exactly where a screenshot should be inserted and what it should show. Replace each placeholder image with the corresponding screenshot once captured.

5.1 Tool Banner & Help Output

Run `python3 reconwraith.py --help` (or `reconwraith --help` if installed to PATH) and capture the full terminal output showing the ASCII banner and argument list.



```
(venv)-(root@kali)-[~/afsataj/ReconWraith]
# python3 reconwraith.py --help
usage: reconwraith [-h] [-o OUTPUT] [--format {txt,json,both}] [--skip-whois] [--skip-dns] [--skip-ports] [--skip-subdomains] [--skip-http]
                  [--ports PORTS] [--wordlist WORDLIST] [--full] [-y] [-v]
                  target

ReconWraith - Automated Reconnaissance Tool for authorized security testing.

positional arguments:
  target                Target domain or IP address (e.g. example.com or 10.0.0.5)

options:
  -h, --help            show this help message and exit
  -o, --output OUTPUT    Output file prefix (default: reconwraith_<target>_<timestamp>)
  --format {txt,json,both}
                        Report format (default: both)
  --skip-whois          Skip WHOIS lookup
  --skip-dns            Skip DNS enumeration
  --skip-ports          Skip port scanning
  --skip-subdomains     Skip subdomain enumeration
  --skip-http           Skip HTTP header grabbing
  --ports PORTS         Comma-separated custom port list, e.g. 22,80,443
  --wordlist WORDLIST    Path to a custom subdomain wordlist file
  --full                Run every module (equivalent to not skipping anything, plus tech-fingerprint & robots)
  -y, --yes             Skip the authorization confirmation prompt (use in automated/CI contexts only)
  -v, --version          show program's version number and exit

(venv)-(root@kali)-[~/afsataj/ReconWraith]
```

5.2 Authorization Prompt

Run a scan without the -y flag and capture the legal/authorization notice and confirmation prompt before typing 'yes'.

```
(venv)-(root@kali)-[~/afsataj/ReconWraith]
# python3 reconwraith.py sbs.du.ac.in

ReconWraith
Automated Reconnaissance Tool | v1.0.0

LEGAL / AUTHORIZATION NOTICE
Only run ReconWraith against systems you own or are explicitly
authorized in writing to test. Unauthorized scanning may be illegal
in your jurisdiction.

Type 'yes' to confirm you are authorized to scan 'sbs.du.ac.in':
```

5.3 Completed Scan Summary

```
python3 reconwraith.py sbs.du.ac.in

ReconWraith
Automated Reconnaissance Tool | v1.0.0

LEGAL / AUTHORIZATION NOTICE
Only run ReconWraith against systems you own or are explicitly
authorized in writing to test. Unauthorized scanning may be illegal
in your jurisdiction.

Type 'yes' to confirm you are authorized to scan 'sbs.du.ac.in': yes
[16:41:57] [INFO] Target: sbs.du.ac.in | Resolved IP: 34.100.172.90
[16:41:57] [INFO] Running WHOIS lookup...
[16:41:58] [OK] WHOIS lookup complete.
[16:41:58] [INFO] Enumerating DNS records...
[16:41:59] [OK] DNS enumeration complete.
[16:41:59] [INFO] Grabbing HTTP/HTTPS banners & headers...
[16:42:02] [OK] HTTP header grab complete.
[16:42:02] [INFO] Fingerprinting web technologies...
[16:42:02] [OK] Technology fingerprinting complete.
[16:42:02] [INFO] nmap found - running nmap service/version scan...
[16:42:18] [OK] nmap scan complete.
[16:42:18] [INFO] Enumerating subdomains (crt.sh + brute force)...
[16:42:29] [WARN] crt.sh lookup failed or timed out: The read operation timed out
[16:42:30] [OK] Subdomain enumeration complete - 1 live subdomain(s) found.
[16:42:30] [OK] Text report saved to reconwraith_sbs.du.ac.in_1783809750.txt
[16:42:30] [OK] JSON report saved to reconwraith_sbs.du.ac.in_1783809750.json
[16:42:30] [OK] Scan complete.
```

5.4 Sample .txt Report

Full command reference and examples are provided in the accompanying USAGE.md file distributed with this project. Key flags are summarized below.

Flag	Purpose
-o, --output	Output file prefix (default: auto-generated with target + timestamp)
--format	txt / json / both (default: both)
--skip-whois / --skip-dns / --skip-ports / --skip-subdomains / --skip-http	Disable individual modules
--ports	Comma-separated custom port list
--wordlist	Path to a custom subdomain wordlist
--full	Enables extra modules (robots.txt / sitemap.xml)
-y, --yes	Bypass the interactive authorization prompt (automation only)

7. Known Limitations

- Subdomain brute force relies on a modest built-in wordlist by default; supply `--wordlist` for larger coverage.
- Without `nmap` installed, port scanning is limited to TCP connect scans on a curated port list (no UDP, no full 65535-port sweep).
- `crt.sh` (certificate transparency lookup) availability depends on that third-party service's uptime and rate limits.
- The tool does not perform active exploitation, vulnerability scanning, or brute-force credential attacks — it is reconnaissance-only by design.

8. Future Enhancements

- Optional HTML report output with charts for open ports and subdomain counts.
- Plug-in architecture for community-contributed recon modules.
- Rate-limiting / stealth timing controls for large-scope engagements.
- Integration with Shodan/Censys APIs for passive port/service data (API key optional).

9. Conclusion

ReconWraith consolidates the manual, repetitive first steps of an authorized security assessment into a single, auditable command with built-in fallbacks and a clear authorization checkpoint. The screenshots referenced in Section 5 should be captured against a lab or explicitly authorized target and inserted in place of each placeholder to complete this report for internal or client-facing distribution.